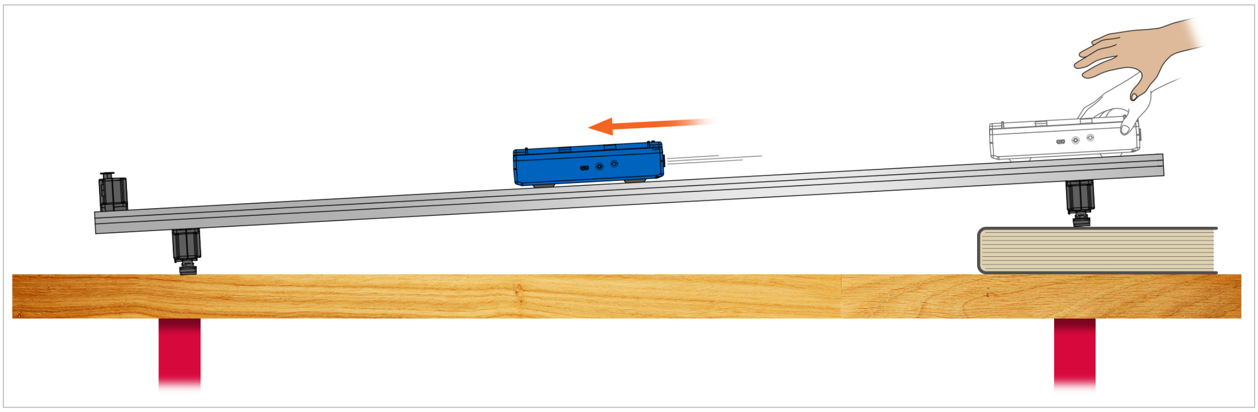
Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • GCSE Physics

Kinematics L3: Acceleration Down a Ramp



Release the cart from the top and let it run down. Equipment illustration courtesy of PASCO scientific.

Set up

1. Keep the same cart and the same laptop graphs as Experiment 1 — position and velocity, both against time.
2. Put **one book** under one end of the track. A small slope is enough. A steep one just means the cart is gone before the laptop has drawn anything.
3. Fit the **end stop at the bottom** of the track. The cart must have something to hit that is not the floor.
4. Turn the cart so the $+X$ **arrow points down the slope**. Rolling down is now the positive direction.
5. Practice the release once, with no recording: hold the cart at the top, **let go without pushing**, and let it run into the end stop.

Let go — do not push. The whole experiment is about what gravity does to the cart on its own. A push at the start puts a jump in the graph that is nothing to do with the ramp, and you will be able to see it.

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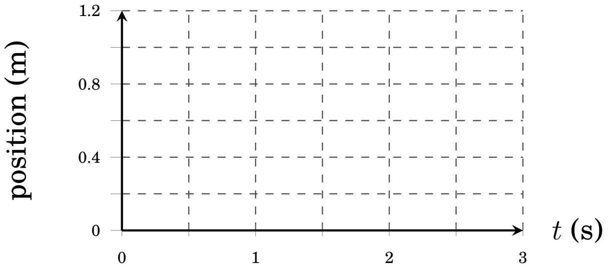
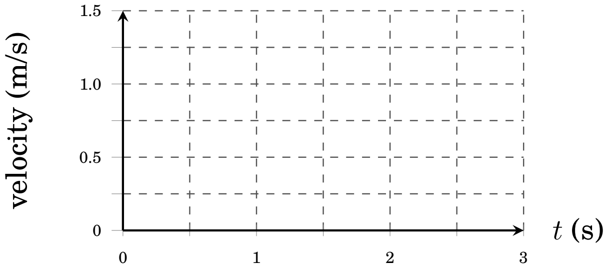
The runs

Run each one, then sketch what the screen showed. Sketch what you actually see, not what you think it ought to look like.

Run 1. One book. Hold at the top, start recording, let go.

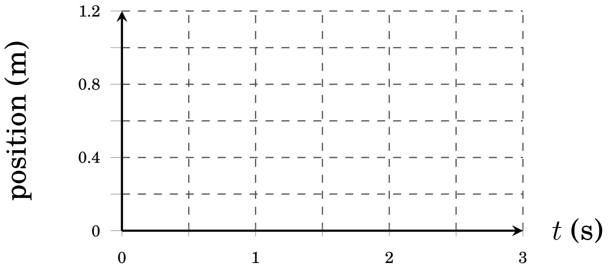
Sketch the position–time graph and the velocity–time graph.

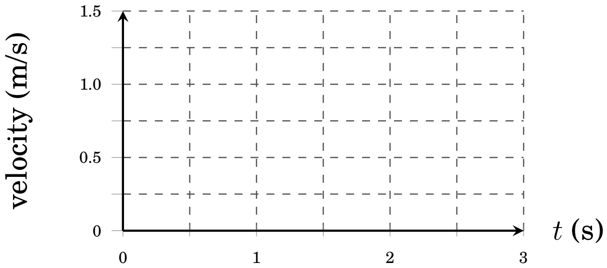




Run 2. Two books — a steeper ramp. Same release.

Sketch the position–time graph and the velocity–time graph.





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Your results

Key concept

Use SPARKvue's **Linear Fit** on the **velocity–time** line. It gives you

$$y = mx + b$$

where m is the **slope** and b is the **y-intercept** — the velocity the cart had at the very start of the run.

In Experiment 1 every velocity–time slope came out as zero. **Watch what m does here.**

Fit only the part where the cart is running freely — leave out your hand at the start and the end stop at the finish.

Run	**Slope m of v–t**	**y-intercept b of v–t / $\mathbf{m\,s^{-1}}$**
1 one book		
2 two books		
3 up and back		

Now compare the two velocity lines from runs 1 and 2.

1) **Describe** what *both* velocity lines have in common, and how they differ.

2) **Explain** what the position–time line was doing in run 1, and why it was not straight.

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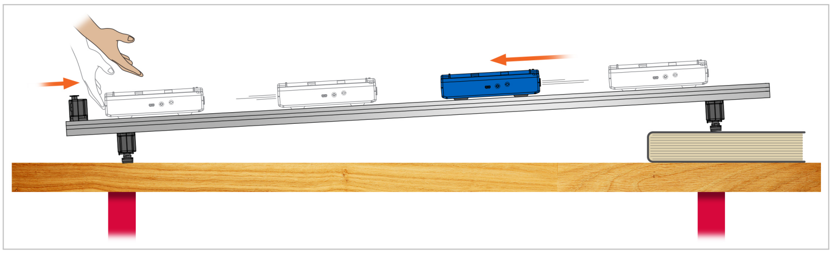
Run 3 — up the ramp, and back down



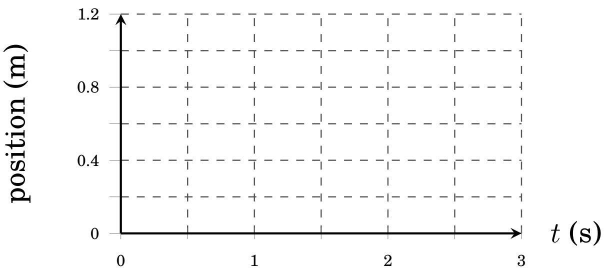
Push it up *the slope and let it come back. Illustration: PASCO scientific.*

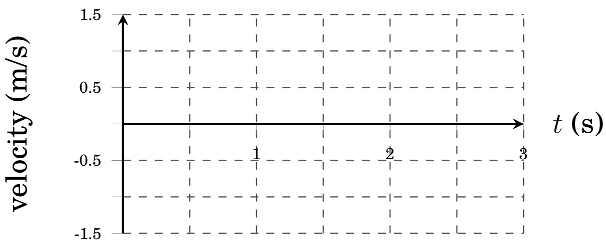
Put the ramp back to **one book**. Start recording, then push the cart **up** the slope from the bottom and let it roll all the way back down. Stop recording before it reaches the end stop. **Catch it.**

For this run only, turn the cart so $+X$ **points *up* the slope** — up is now positive, so your position graph climbs from zero and comes back to it.

Run 3. Pushed up the slope, slows, stops, comes back down.

Sketch the position–time graph and the velocity–time graph.





1) Identify the moment the cart was momentarily still. What was the velocity line doing at exactly that moment?

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2) Explain why the velocity line crosses the zero line in this run, when it never did in Experiment 1.

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3) Describe the shape of the velocity line over the whole run, in one sentence.

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